

CORPORATE TRAINING PROGRAMME

Microsoft Power Automate

Workflow Automation for Workplace Productivity

Trainer:

Fakhrul Syahmi

Data Consultant & AI Trainer

mufasyahcons@gmail.com

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About the Trainer



Dr. Fakhrul Syahmi
Data Consultant
& AI Trainer

Professional Background

Fakhrul Syahmi is a certified Data Consultant and Corporate Trainer specializing in Microsoft Power Platform, Business Intelligence, and AI-powered workplace productivity. With extensive experience delivering training to MNCs, government bodies, and higher education institutions, he brings practical, hands-on expertise to every session.

Core Expertise

- Microsoft Power Automate, Power BI, Power Apps, Power Platform
- Microsoft Excel (Advanced), Power Query, Power Pivot, DAX
- AI Tools: Microsoft Copilot, ChatGPT, Google Gemini
- Data Science, Machine Learning, LDA Topic Modelling
- Corporate Training for Perodua, Bank Rakyat, TNB, Teleperformance, HELP University

Academic Qualifications

- Master of Data Science
- Bachelor's Degree in Economics
- Advanced Studies in Islamic Studies and Arabic Language

✉ mufasyahcons@gmail.com

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Programme Learning Objectives

Upon completion of this training programme, participants will be able to:

1. Understand the fundamentals of workflow automation and the Microsoft Power Platform ecosystem.
2. Navigate the Power Automate interface and create cloud-based automated flows from scratch.
3. Build automated email notifications, file management, and SharePoint integration flows.
4. Implement conditional logic, variables, expressions, and loops within complex flows.
5. Design and deploy multi-stage approval workflows for HR, Finance, and Operations use cases.
6. Integrate Power Automate with Microsoft 365 applications including Outlook, Teams, Excel, and SharePoint.
7. Utilise AI Builder features for intelligent document processing, OCR, and sentiment analysis.
8. Automate repetitive desktop and legacy system tasks using Power Automate Desktop (RPA).
9. Apply governance best practices including naming conventions, security, and environment management.
10. Analyse real-world business case studies and replicate end-to-end automation architectures.
11. Complete guided lab exercises and build a full business automation solution as a final project.



TRAINER NOTE

Duration: 2-day corporate training (approximately 16 hours of instructional content)

Delivery mode: In-person or virtual (Microsoft Teams recommended for remote delivery)

Prerequisites: Basic Microsoft 365 familiarity recommended but not mandatory

All participants should have access to a Microsoft 365 account with Power Automate enabled

MODULE 1

Introduction to Power Automate

Understanding Workflow Automation & the Microsoft Power Platform

Module 1 — Introduction to Power Automate

This module establishes the foundational understanding of automation, digital transformation, and where Microsoft Power Automate fits within the broader Microsoft Power Platform. Participants will explore real-world business use cases and understand the value automation brings to modern workplaces.

1.1 What is Workflow Automation?

Workflow automation is the process of using technology to perform repetitive, rule-based tasks with minimal or no human intervention. Rather than manually executing each step of a business process, automation tools execute predefined sequences of actions triggered by specific events or conditions.

In a traditional workplace, employees often spend significant portions of their working day on tasks that are predictable, repetitive, and do not require strategic thinking — such as:

- Sending follow-up emails after a meeting
- Copying data from one spreadsheet to another
- Getting approvals via email chains
- Downloading reports and saving them to shared drives
- Sending reminders for deadlines

Workflow automation replaces these manual steps with automated sequences, freeing employees to focus on higher-value work such as analysis, decision-making, and customer engagement.



REAL-WORLD SCENARIO

SCENARIO: Sarah is an HR Executive at a manufacturing company. Every time a new employee joins, she manually sends a welcome email, creates a SharePoint folder, adds the employee to Teams, and updates an Excel tracking sheet. This process takes approximately 45 minutes per new hire.

With Power Automate: When a new employee record is submitted via Microsoft Forms, all four tasks are triggered automatically — completed in under 2 minutes with zero manual effort.

Estimated Annual Saving: If 50 employees are onboarded per year — Sarah reclaims over 35 hours of productive working time.

1.2 Benefits of Automation in the Workplace

Organisations that adopt workflow automation consistently report measurable improvements across multiple performance dimensions:

Benefit	Business Impact
Time Savings	Eliminates manual, repetitive tasks — employees reclaim hours each week
Reduced Errors	Eliminates human data-entry errors, ensuring data consistency and accuracy
Faster Turnaround	Approval cycles that took days via email can complete in minutes
Cost Reduction	Lower operational costs through reduced manual labour and fewer errors
Scalability	Automated processes handle increased volume without additional staff
Compliance	Automated audit trails ensure processes are documented and traceable
Employee Satisfaction	Staff focus on meaningful work rather than administrative burdens
24/7 Operation	Automated workflows run outside business hours without supervision

QUICK TIP

When presenting automation ROI to leadership, calculate: (Time saved per task × Number of occurrences per month × Hourly staff cost). Even a 5-minute saving per daily task yields over 20 hours saved per year per employee.

1.3 Digital Transformation & the Role of Automation

Digital transformation is the process of integrating digital technology into all areas of a business, fundamentally changing how it operates and delivers value. Workflow automation is a cornerstone of digital transformation strategy.

Microsoft identifies four pillars of digital transformation, and Power Automate contributes to all of them:

- Empower Employees: Automation tools that non-technical staff can use independently (citizen development)
- Engage Customers: Faster responses, automated follow-ups, and consistent service delivery
- Optimise Operations: Streamlined processes, reduced bottlenecks, and data-driven decision making
- Transform Products: Intelligent, connected digital services and products powered by automation and AI

 TRAINER DEMONSTRATION

Show participants the Microsoft Digital Transformation documentation at [microsoft.com](https://microsoft.com/digital)
Ask: 'Which department in your organisation has the most repetitive manual tasks?'
Collect responses — these become potential automation use cases for Module 11 labs

1.4 Microsoft Power Platform Overview

Microsoft Power Platform is a suite of low-code/no-code tools that enables organisations to analyse data, build solutions, automate processes, and create virtual agents. Power Automate is one of five core components:

Component	Icon	Primary Purpose	Typical User
Power BI		Data analytics and interactive dashboards	Analysts, Managers, Executives
Power Apps		Build custom business applications	Operations, IT, Business teams
Power Automate		Workflow and process automation	All business users
Power Pages		Low-code external-facing websites	Web teams, IT departments
Copilot Studio		Build AI-powered virtual agents/chatbots	Customer service, HR, IT

These components are designed to work seamlessly together. For example, a Power App can collect data that triggers a Power Automate flow, which updates a SharePoint list, which refreshes a Power BI dashboard — creating an end-to-end intelligent business solution.

 INSERT SCREENSHOT HERE

Caption: Microsoft Power Platform Admin Centre — showing all five platform components with navigation menu

1.5 Microsoft Power Automate Overview

Microsoft Power Automate (formerly known as Microsoft Flow) is a cloud-based service that enables users to create automated workflows between applications and services. It can automate repetitive tasks and business processes across Microsoft 365 and hundreds of third-party services.

Power Automate supports three primary flow types:

Flow Type	Description & Use Cases
Cloud Flows	Run entirely in the cloud; connect to online services like SharePoint, Outlook, Teams, Salesforce, and 1000+ connectors. Best for

	email automation, approval workflows, and data synchronisation.
Desktop Flows (RPA)	Automate tasks on your Windows desktop, including legacy applications, web browsers, and local files. No API required — records and replays UI interactions.
Business Process Flows	Guide users through standardised multi-step processes within Dynamics 365 or Power Apps; ensure data quality and process consistency.

1.6 Real Business Use Cases

Power Automate is deployed across virtually every industry and business function. The following examples represent common automation scenarios that participants may recognise from their own organisations:

Human Resources

- Employee onboarding — automatic folder creation, welcome emails, Teams channel additions
- Leave approval workflows with multi-level authorisation
- HR document request and approval routing
- New joiner equipment request management

Finance & Procurement

- Invoice approval routing with escalation rules
- Purchase order request and approval workflows
- Automated expense report notifications
- Month-end reporting data consolidation

Operations & Administration

- Meeting room booking confirmation and reminder flows
- Automated daily/weekly reports from SharePoint to Teams or email
- Customer feedback form → ticket creation pipeline
- Supplier onboarding document collection and verification

DISCUSSION QUESTIONS

- Q1: What are the top 3 most time-consuming manual tasks in your current role?
- Q2: Have you ever sent the same type of email repeatedly? What triggered it?
- Q3: Where do delays typically occur in your team's approval processes?

1.7 Licensing Overview

Power Automate is available under several licensing tiers. Understanding licensing ensures participants know what features are available in their organisation:

Plan	Description
Microsoft 365 Included	Basic cloud flows using standard connectors included with most Microsoft 365 subscriptions (Business Basic, Business Standard, E1, E3, E5)
Power Automate Premium (per user)	Includes premium connectors, attended desktop flows, and process mining. Required for advanced automation.
Power Automate Process	For unattended desktop RPA flows running without a logged-in user — for server-based automation.
Power Automate Premium per Flow	Organisation-wide unlimited flow runs for a specific flow, suitable for high-volume business processes.

IMPORTANT NOTE

Participants should check with their IT administrator to confirm which Power Automate licence is assigned to their Microsoft 365 account before attempting premium connector exercises.

Most exercises in this training use standard connectors available to all Microsoft 365 users.

1.8 Cloud Flows vs Desktop Flows

Feature	Cloud Flows vs Desktop Flows (RPA)
Technology	Cloud Flows: API-based connections Desktop Flows: UI/screen automation
Connectivity	Cloud Flows: Online services only Desktop Flows: Local apps, legacy systems, web browsers
Setup	Cloud Flows: Browser-based designer Desktop Flows: Power Automate Desktop app (Windows)
Trigger	Cloud Flows: Event-driven or scheduled Desktop Flows: Called by cloud flows or manual
Use when	Cloud Flows: Systems have APIs Desktop Flows: No API available (e.g., SAP, older ERP)
Examples	Cloud Flows: Email routing, SharePoint sync Desktop Flows: Filling legacy web forms, ERP data entry

MODULE 1 SUMMARY

- ✓ Workflow automation replaces repetitive manual tasks with rule-based automated sequences
- ✓ Benefits include time savings, error reduction, cost efficiency, and improved compliance
- ✓ Microsoft Power Platform is a suite of 5 tools: Power BI, Apps, Automate, Pages, Copilot Studio
- ✓ Power Automate has 3 flow types: Cloud Flows, Desktop Flows (RPA), and Business Process Flows
- ✓ Real use cases span HR, Finance, Operations, and Administration across all industries
- ✓ Licensing ranges from Microsoft 365 included plans to dedicated Premium and Process plans

KNOWLEDGE CHECK

- Q1: Name two benefits of workflow automation in a corporate environment.
- Q2: What is the difference between a Cloud Flow and a Desktop Flow?
- Q3: Which Microsoft Power Platform tool is primarily used for creating chatbots?
- Q4: True or False — Power Automate was previously called Microsoft Flow.

ANSWERS: Q1: Any two from — time savings, error reduction, cost reduction, scalability, compliance, 24/7 operation | Q2: Cloud flows use APIs to connect online services; Desktop flows automate local/legacy apps via UI | Q3: Copilot Studio (formerly Power Virtual Agents) | Q4: TRUE

MODULE 2

Getting Started with Power Automate

Interface, Connectors, Triggers & Your First Flow

Module 2 — Getting Started with Power Automate

In this module, participants will log into Power Automate, explore the interface, understand the core building blocks — connectors, triggers, and actions — and create their very first automated cloud flow.

2.1 Accessing Power Automate

Power Automate can be accessed through multiple entry points:

12. Open a browser and navigate to: <https://make.powerautomate.com>
13. Alternatively, from Microsoft 365 portal (office.com) → Click the App Launcher (grid icon) → Power Automate
14. From Microsoft Teams: Apps → Search 'Power Automate' → Add to Teams
15. Download Power Automate Desktop from the Microsoft Store (for RPA/Desktop flows)



INSERT SCREENSHOT HERE

Caption: Power Automate login page at make.powerautomate.com — showing Microsoft account sign-in



TRAINER NOTE

Ask all participants to log in to Power Automate now. Verify each participant can see the Home dashboard.

If participants cannot log in, ask them to confirm their Microsoft 365 licence with their IT department.

2.2 Interface Overview

The Power Automate interface is organised into a clean, intuitive navigation structure:

Navigation Section	Description
Home	Dashboard with recommended templates, recent flows, and quick actions
My Flows	List of all flows you have created or shared with you

Create	Central hub for creating new flows (automated, scheduled, instant, desktop, AI)
Templates	Pre-built flow templates for common automation scenarios
Connectors	Browse 1,000+ available connectors to services and applications
Process Mining	Analyse existing process data to identify automation opportunities
Solutions	Package flows and components for deployment across environments
Learn	Access Microsoft Learn tutorials, documentation, and guided learning



INSERT SCREENSHOT HERE

Caption: Power Automate left navigation panel — highlighting Home, My Flows, Create, Templates, Connectors

2.3 Understanding Environments

Environments in Power Automate are containers that separate flows, data, and resources. Understanding environments is essential for managing flows correctly in a corporate setting.

- Default Environment: Automatically created for your organisation; used by most individual users
- Production Environment: Managed by IT/Admins; used for business-critical flows deployed company-wide
- Sandbox/Development Environment: Used for testing and development before promoting to production
- Trial Environment: Available when evaluating premium features



IMPORTANT NOTE

In a corporate setting, always confirm with your IT administrator which environment you should be working in.
Flows created in the Default environment may not be accessible to other environments or vice versa.

2.4 Connectors — Connecting the Digital World

Connectors are the bridges that allow Power Automate to communicate with external applications and services. There are over 1,000 connectors available, spanning Microsoft products, third-party SaaS tools, and custom APIs.

Connector Type	Description & Examples
----------------	------------------------

Standard Connectors	Available to all Microsoft 365 users — includes SharePoint, Outlook, Teams, Excel, OneDrive, Forms, Planner, To Do
Premium Connectors	Require Power Automate Premium licence — includes Salesforce, ServiceNow, SAP, SQL Server, HTTP webhooks, Adobe Sign
Custom Connectors	Built by your IT team to connect to proprietary internal APIs and systems
On-Premises Gateway	Allows cloud flows to securely access on-premises data sources like SQL Server, Oracle, or file shares

QUICK TIP

Before designing a flow, always check whether a connector exists for each application you want to automate. Search at: make.powerautomate.com/connectors

2.5 Triggers and Actions — The Building Blocks

Every Power Automate flow is structured around two fundamental concepts:

Triggers

A trigger is the event that starts the flow. There is always exactly one trigger per flow. Triggers can be:

- Event-based: 'When a new email arrives in Outlook', 'When a file is created in SharePoint'
- Scheduled: 'Every Monday at 8:00 AM', 'On the 1st of every month'
- Instant/Manual: 'When I press a button', 'When called by another flow'

Actions

Actions are the steps the flow executes after the trigger fires. A flow can have unlimited actions. Examples:

- Send an email via Outlook
- Create a file in OneDrive or SharePoint
- Post a message in Microsoft Teams
- Update a row in an Excel table
- Start an approval process
- Send an HTTP request to an API

Concept	Analogy
Trigger	Like a doorbell — something happens that starts the process
Action	Like what you do after the doorbell rings — each step you take

Condition	Like deciding whether to answer the door — 'if this, do that'
Loop	Like checking each package in a delivery — repeating for every item

2.6 Creating Your First Flow — Step-by-Step

In this guided exercise, we will create a simple automated flow that sends an email notification whenever a new file is added to a OneDrive folder.

Step-by-Step Tutorial: OneDrive File Alert

16. Go to make.powerautomate.com and sign in with your Microsoft 365 account
17. Click 'Create' from the left navigation panel
18. Select 'Automated cloud flow'
19. In the 'Build an automated cloud flow' dialog: Enter Flow name: 'OneDrive New File Alert'
20. In the search box, type 'OneDrive' and select: 'When a file is created (OneDrive for Business)'
21. Click 'Create'
22. In the trigger configuration: Set Folder to '/Training Files' (or create this folder in your OneDrive)
23. Click '+ New step'
24. Search for 'Send an email' and select: 'Send an email (V2)' from Office 365 Outlook
25. Configure the email action: To: your own email address | Subject: 'New file added: [File name]' (use dynamic content) | Body: 'A new file has been added to your OneDrive. File: [File name] | Created: [Created time]'
26. Click 'Save' (top right corner)
27. Click 'Test' → select 'Manually' → click 'Test'
28. Upload a test file to your OneDrive Training Files folder
29. Wait 1-2 minutes and check your email inbox for the notification



INSERT SCREENSHOT HERE

Caption: Flow designer canvas showing OneDrive trigger connected to Outlook Send Email action



PARTICIPANT ACTIVITY

ACTIVITY 2A: Create your first flow as described above.

Time allowed: 20 minutes

Success criteria: You receive an email notification when a file is added to OneDrive

If stuck: Raise your hand — the trainer will assist

2.7 Using Templates

Power Automate provides hundreds of pre-built templates to help you get started quickly. Templates are complete flow blueprints that you can customise for your specific needs.

To use a template:

30. Click 'Templates' in the left navigation
31. Browse by category or search by keyword (e.g., 'approval', 'Teams notification', 'SharePoint')
32. Click any template to preview it
33. Click 'Use this template'
34. Sign in to the required connectors and modify settings as needed
35. Click 'Continue' and 'Save'

QUICK TIP

Recommended starter templates: 'Send a customised email when a new SharePoint list item is added', 'Get a notification when you receive an email from your boss', 'Save Office 365 email attachments to OneDrive'

2.8 Saving and Testing Flows

Best practices for saving and testing:

- Always name your flow clearly before saving (use the naming convention discussed in Module 9)
- Click 'Save' frequently while building — Power Automate does not auto-save continuously
- Use 'Test' → 'Manually' to run a flow and verify it works end-to-end
- Use 'Test' → 'With a recently triggered flow' to reuse the last successful trigger data
- After testing, review the 'Flow checker' panel for warnings and errors
- Check 'My Flows' → Click flow name → 'Run history' to see past execution logs

COMMON MISTAKES

Forgetting to save the flow before testing — the Test button won't work on unsaved changes
Not connecting all required connectors — a red warning icon appears on disconnected steps
Testing with incorrect trigger conditions — ensure the trigger event actually fires during testing
Leaving flows in 'Turned Off' state — check the flow status indicator in My Flows

MODULE 2 SUMMARY

- ✓ Access Power Automate via make.powerautomate.com or via the Microsoft 365 app launcher
- ✓ The interface includes: Home, My Flows, Create, Templates, Connectors, Solutions
- ✓ Environments separate flows for different purposes: Default, Production, Sandbox, Trial
- ✓ Connectors link Power Automate to 1,000+ external services (Standard and Premium tiers)
- ✓ Every flow has one Trigger and one or more Actions
- ✓ Templates provide pre-built starting points for common automation scenarios
- ✓ Always save and test your flows before sharing or deploying to production

 KNOWLEDGE CHECK

- Q1: What is the URL to access Power Automate in a browser?
- Q2: What is the difference between a Standard and a Premium connector?
- Q3: Every Power Automate flow must have at least one _____ and one _____.
- Q4: Name three ways to trigger a Power Automate flow.

ANSWERS: Q1: make.powerautomate.com | Q2: Standard connectors are included with Microsoft 365; Premium require an additional paid licence | Q3: Trigger, Action | Q4: Any three from — event-based (email received, file created), scheduled (daily/weekly), instant/manual (button press, called by another flow)

MODULE 3

Building Basic Automated Flows

Email, SharePoint, Teams, Excel & OneDrive Integration

Module 3 — Building Basic Automated Flows

This module provides hands-on experience building practical automated flows using the most common Microsoft 365 services. Participants will learn to automate email notifications, save attachments, integrate with SharePoint, send Teams alerts, update Excel data, and generate automated reports.

3.1 Email Notification Automation

One of the most common automation use cases is triggering actions based on email events. The Outlook connector provides powerful triggers and actions for email automation.

Common Email Automation Scenarios

- Send an automatic reply when an email arrives from a specific sender
- Forward high-priority emails to your manager automatically
- Create a Teams message when an email with a specific subject arrives
- Log incoming customer emails to a SharePoint list for tracking

Tutorial: Auto-Notify Team When Email Arrives

36. Create → Automated cloud flow → Trigger: 'When a new email arrives (V3)' — Outlook
37. Configure trigger: Folder: Inbox | Subject filter: 'Project Update' | Importance: High
38. Add action: 'Post a message in a chat or channel' — Microsoft Teams
39. Configure Teams action: Post in: Channel | Team: [Your team] | Channel: General
40. Message: 'New high-priority project email received from: [From] Subject: [Subject]'
41. Save and test by sending a test email with 'Project Update' in the subject

**INSERT SCREENSHOT HERE**

Caption: Email trigger configuration panel — showing Subject Filter, Folder, and Importance settings

3.2 File Approval Workflows

SharePoint is commonly used as a document management system. Power Automate can trigger workflows when documents are uploaded, modified, or submitted for review.

A simple file approval flow works as follows:

42. Employee uploads a document to a SharePoint document library
43. Flow is triggered automatically
44. Flow sends an approval request to the line manager via email/Teams
45. Manager approves or rejects
46. Uploader is notified of the decision
47. Document is moved to 'Approved' or 'Rejected' folder accordingly

TRAINER DEMONSTRATION

Demonstrate a simple approval flow using a SharePoint document library
Show how dynamic content pulls the file name and uploader's name into the approval notification
Highlight the approval email that appears in Outlook with Approve/Reject buttons

3.3 Saving Email Attachments Automatically

A highly popular automation is automatically saving email attachments to OneDrive or SharePoint — eliminating the need to manually download and organise files.

Tutorial: Auto-Save Email Attachments to SharePoint

48. Create → Automated cloud flow
49. Trigger: 'When a new email arrives (V3)' — configure: Has Attachments = Yes
50. Add action: 'Apply to each' → Dynamic content: Attachments
51. Inside the loop, add: 'Create file' — SharePoint
52. Configure: Site Address: [Your SharePoint site] | Folder: /Email Attachments/2026
53. File Name: [Attachment Name] | File Content: [Attachment Content]
54. Save and test by sending an email with a PDF attachment to yourself

INSERT SCREENSHOT HERE

Caption: Flow showing Apply to Each loop with SharePoint Create File action — dynamic content panel visible

QUICK TIP

Add a Condition before creating the file to check if the attachment is a PDF or specific file type, preventing unwanted file types (e.g., email signatures as .png) from being saved.

3.4 SharePoint Integration

SharePoint is the backbone of many corporate automation scenarios. Power Automate integrates deeply with SharePoint lists and document libraries.

Trigger

Example Use Case

When an item is created	New task notification, onboarding trigger, ticket creation alert
When an item is modified	Change approval workflow, audit log update, notification to owner
For a selected item	On-demand flow triggered from SharePoint list view
When a file is created/modified	Document review workflow, attachment processing

Action	Example Use Case
Get items	Retrieve a list of all pending requests from a SharePoint list
Create item	Log automation results, create new task records, register new entries
Update item	Mark items as Approved/Rejected, update status fields
Delete item	Archive or remove processed records
Get file content	Read document contents for processing or forwarding

3.5 Teams Notification Automation

Microsoft Teams is the central collaboration hub for most organisations. Power Automate can post messages, create channels, and send adaptive cards directly to Teams.

- 'Post a message in a chat or channel' — posts to a team channel or group chat
- 'Post an adaptive card and wait for a response' — interactive card with input fields
- 'Create a channel' — programmatically create Teams channels
- 'Send a message to a user in Microsoft Teams' — direct 1:1 message



REAL-WORLD SCENARIO

SCENARIO: Azman manages a customer support team. Every time a customer submits a complaint form via Microsoft Forms, the team should be notified in their Teams channel with the customer name, complaint category, and priority level.

SOLUTION: Forms submission → Power Automate triggers → Extract form responses → Post adaptive card to Teams #support channel with customer details → Assign to available agent using Teams message reply.

3.6 Excel Automation

Power Automate can read from and write to Excel files stored in OneDrive or SharePoint — enabling automated data updates without opening Excel manually.

⚠ IMPORTANT NOTE

Excel files must be stored in OneDrive or SharePoint (not locally on your computer) for Power Automate to access them.

The Excel file must contain a properly formatted Table (Insert → Table in Excel) for Power Automate to interact with rows.

Power Automate works with Excel Online — it does not support the legacy .xls format.

Excel Action	Use Case
List rows in a table	Read all data from an Excel table — e.g., retrieve all employee records
Add a row into a table	Append new data — e.g., log each form submission into a tracking spreadsheet
Update a row	Modify existing data — e.g., mark an order as 'Shipped'
Get a row	Retrieve a specific record by row ID — e.g., look up a client record
Delete a row	Remove records — e.g., archive processed entries
Run script	Execute Office Script (JavaScript) for complex Excel transformations

3.7 OneDrive Integration

OneDrive for Business serves as personal cloud storage for each Microsoft 365 user. Power Automate can monitor folders, create files, copy, move, and delete files in OneDrive.

Common OneDrive Automation Scenarios

- Copy files between folders based on naming conventions or metadata
- Automatically organise downloaded files by year/month/category
- Sync files between OneDrive and SharePoint
- Create a daily folder structure for report storage
- Archive files older than a specified date to an archive folder

3.8 Automated Reporting Examples

Power Automate can generate and send scheduled reports by combining triggers (Schedule), data retrieval (SharePoint, Excel), and communication (Outlook, Teams).

Sample: Weekly Sales Summary Email

55. Trigger: Recurrence — Every Monday at 8:00 AM
56. Action: Get items from SharePoint 'Weekly Sales' list → filter by current week
57. Action: Create HTML table from the retrieved items
58. Action: Send email with the HTML table embedded in the body



MODULE 3 SUMMARY

- ✓ Email automation uses Outlook triggers — filter by sender, subject, importance, or attachments
- ✓ Saving email attachments uses 'Apply to each' loop with SharePoint/OneDrive Create File action
- ✓ SharePoint integration offers rich triggers (item created/modified) and actions (get/create/update items)
- ✓ Teams automation enables channel posts, direct messages, and interactive adaptive cards
- ✓ Excel in Power Automate requires: file in OneDrive/SharePoint + formatted Table in the spreadsheet
- ✓ Automated reports combine Recurrence triggers with data retrieval and email/Teams delivery

MODULE 4

Conditions, Variables & Logic

Building Intelligent, Decision-Driven Automation

Module 4 — Conditions, Variables & Logic

Real-world automation requires decision-making. This module covers the logical building blocks that allow flows to make decisions, store data in memory, loop through collections, handle errors gracefully, and process complex business rules.

4.1 Conditions (IF Statements)

A Condition in Power Automate works like an IF statement — it evaluates whether something is true or false, and takes different paths based on the result.

Structure: IF [value] [operator] [comparison value] → THEN YES / NO

Condition Operator	Example
is equal to	Status is equal to 'Approved'
is not equal to	Department is not equal to 'IT'
is greater than	Amount is greater than 5000
is less than	Leave days is less than 3
contains	Email subject contains 'Invoice'
starts with	File name starts with 'REPORT_'
is empty	Approver field is empty → assign default approver

Tutorial: Conditional Email Routing

- 59. Create an automated flow triggered by a SharePoint list item being created
- 60. Add a 'Condition' action
- 61. Left value: Dynamic content → Priority field | Operator: is equal to | Right value: High
- 62. In the YES branch: Send email with subject 'URGENT: High-priority request received'
- 63. In the NO branch: Send email with subject 'New request submitted — standard processing'

 INSERT SCREENSHOT HERE

Caption: Condition action in flow designer — showing YES (green) and NO (red) branches

4.2 Variables — Storing Data in Memory

Variables allow you to store, track, and update values as your flow runs. They are essential for counting iterations, building dynamic content, and accumulating data.

Variable Type	Description & Use Case
String	Stores text — e.g., store a formatted email body built across multiple steps
Integer	Stores whole numbers — e.g., count the number of approved items in a loop
Float	Stores decimal numbers — e.g., accumulate total invoice amounts
Boolean	Stores true/false — e.g., track whether any errors occurred
Array	Stores a list of values — e.g., collect names of all approvers
Object	Stores structured data — e.g., store a complete JSON record

Variable Actions in Power Automate

- 'Initialize variable' — declare the variable at the start of the flow (must be done before the variable is used)
- 'Set variable' — replace the variable's value completely
- 'Increment variable' — add a number to an integer or float variable
- 'Decrement variable' — subtract a number from an integer or float variable
- 'Append to string variable' — add text to the end of an existing string variable
- 'Append to array variable' — add an item to an array

IMPORTANT NOTE

'Initialize variable' must ALWAYS be placed before any loops or conditions that use the variable.

Variables initialized inside an Apply to Each loop reset with each iteration — initialize them BEFORE the loop.

4.3 Dynamic Content

Dynamic content is the data passed between steps in a flow. When you configure an action, you can insert dynamic content from previous steps — such as the sender's name, file name, form responses, or SharePoint field values.

 QUICK TIP

Access dynamic content by clicking inside any text field in an action, then clicking the lightning bolt icon or 'Add dynamic content'. The panel shows all available outputs from previous steps in your flow.

4.4 Expressions

Power Automate includes a library of built-in functions called Expressions. These allow you to manipulate data, format text, perform calculations, and work with dates — directly within your flow.

Expression Category	Example Functions
String functions	concat(), toLower(), toUpper(), substring(), replace(), trim(), startsWith(), endsWith()
Date/Time functions	utcNow(), formatDateTime(), addDays(), addHours(), dayOfWeek(), convertTimeZone()
Math functions	add(), sub(), mul(), div(), mod(), max(), min(), rand()
Logical functions	if(), and(), or(), not(), equals(), greater(), less()
Array functions	length(), first(), last(), union(), intersection(), skip(), take()
Conversion functions	string(), int(), float(), bool(), json(), xml(), base64()

Practical Expression Examples

- Get today's date formatted as DD/MM/YYYY: `formatDateTime(utcNow(), 'dd/MM/yyyy')`
- Calculate date 7 days from now: `addDays(utcNow(), 7)`
- Convert email sender to uppercase: `toUpper(triggerOutputs()?['body/from'])`
- Check if a field contains 'urgent': `contains(toLower(items('Apply_to_each')?['Title']), 'urgent')`

4.5 Loops — Apply to Each

The 'Apply to each' action repeats a set of actions for every item in a collection (array). This is the equivalent of a 'for each' loop in programming.

Common Apply to Each Scenarios

- Process each email attachment separately
- Update each row in an Excel table
- Send individual notification emails to each person in a SharePoint list
- Create SharePoint items for each response in a Microsoft Forms submission

Controlling Loop Performance

By default, Apply to Each processes items sequentially. To improve performance:

- Click the three-dot menu (...) on the Apply to Each action
- Select 'Settings'
- Enable 'Concurrency Control' and set degree of parallelism (1-50)

IMPORTANT NOTE

Enabling concurrency (parallel processing) means items are processed simultaneously. This is faster but can cause issues if actions inside the loop depend on each other's outputs.

4.6 Error Handling & Configure Run After

In production flows, errors are inevitable. Power Automate provides built-in error handling to ensure flows behave predictably even when steps fail.

Configure Run After

By default, each action only runs if the previous action succeeded. You can change this behaviour:

- Right-click any action → 'Configure run after'
- Options: is successful | has failed | is skipped | has timed out
- Useful for: sending an error notification if a step fails, skipping optional steps gracefully

Scope Action for Error Handling

Wrap critical steps inside a 'Scope' action, then add error-handling actions after the scope:

64. Add a 'Scope' action and move critical steps inside it
65. After the scope, add a parallel branch set to 'has failed'
66. In the failure branch: send email to admin, log error to SharePoint, post Teams alert

4.7 Debugging Logic

When a flow doesn't behave as expected, use these debugging techniques:

- Check 'Run history' — click the failed flow run to see which step failed and why
- Expand each step to see its inputs and outputs — compare expected vs actual values
- Use 'Compose' action as a debug tool — output any dynamic content or expression to see its value
- Check condition logic — verify that comparison values match exactly (case-sensitive for strings)
- Use 'Terminate' action with 'Failed' status to force-stop a flow and log a custom message

MODULE 4 SUMMARY

- ✓ Conditions (IF statements) enable flows to make decisions based on data values
- ✓ Variables store and update data types: String, Integer, Float, Boolean, Array, Object
- ✓ Dynamic content lets you insert outputs from previous steps into current actions
- ✓ Expressions provide 50+ built-in functions for string, date, math, and logical operations
- ✓ Apply to Each loops repeat actions for every item in a collection

- ✓ Configure Run After enables error handling — define what happens when steps fail or are skipped
- ✓ Use Run History and Compose actions to debug flow logic

MODULE 5

Approval Workflows

Designing Intelligent Approval Systems for HR, Finance & Operations

Module 5 — Approval Workflows

Approval workflows are among the most impactful automation use cases in corporate environments. This module covers designing single-approver, sequential, parallel, and multi-stage approval systems — with real examples from HR and Finance.

5.1 Approval Concepts in Power Automate

Power Automate provides a dedicated Approvals connector that manages the entire approval lifecycle — from request creation to response tracking and notification.

Key approval actions:

- 'Start and wait for an approval' — sends approval request and pauses the flow until a decision is made
- 'Create an approval' — sends the request without waiting (use with parallel processing)
- 'Wait for an approval' — resume flow after a previously created approval completes
- 'Respond to an approval' — used in Teams/Outlook adaptive cards

Approval Type	Description
Approve/Reject - First to respond	Request sent to multiple approvers; the FIRST response determines outcome
Approve/Reject - Everyone must approve	ALL listed approvers must approve; one rejection cancels the request
Custom Responses	Define your own response options (e.g., 'Approve', 'Reject', 'Defer', 'More Info Needed')

5.2 Sequential Approvals

Sequential approvals process approvers one at a time — the next approver only receives the request after the previous approver has responded. This mirrors typical corporate sign-off hierarchies.

Tutorial: 2-Level Sequential Leave Approval

67. Trigger: When a SharePoint list item is created (Leave Request list)
68. Action: Start and wait for approval — Type: Approve/Reject (First to respond)
69. Assigned to: [Line Manager email from SharePoint item]
70. Title: 'Leave Request from [Employee Name]' | Details: '[Leave dates] — [Leave type]'

71. Add Condition: Approval Outcome is equal to 'Approve'
72. YES branch: Add second approval — Assigned to: [HR Manager]
73. Add another condition inside YES: Second approval outcome is Approve
74. YES: Send approval confirmation email to employee | Update SharePoint status to 'Approved'
75. NO (inner): Send rejection email | Update status to 'Rejected by HR'
76. NO (outer): Send rejection email | Update status to 'Rejected by Line Manager'



INSERT SCREENSHOT HERE

Caption: Sequential approval flow canvas — showing two Start and Wait for Approval actions connected in sequence with condition branches

5.3 Parallel Approvals

Parallel approvals send requests to all approvers simultaneously. This is faster than sequential processing and is appropriate when approvers are peers rather than a hierarchy.

Building Parallel Branches

77. After the trigger, add 'Create an approval' (not 'Start and wait') — this sends without pausing
78. Add a parallel branch (click + icon on the connector line between actions)
79. In each branch, add a separate 'Create an approval' for each approver
80. After all parallel branches, add 'Wait for an approval' for each request
81. Evaluate all responses before proceeding

5.4 Multi-Stage Approvals

Multi-stage approvals combine sequential and parallel patterns to model complex real-world governance:

- Stage 1: Line Manager approval (sequential)
- Stage 2: Department Head and Finance approval (parallel — both must approve)
- Stage 3: HR Director final approval (sequential)



REAL-WORLD SCENARIO

SCENARIO: A staff purchase request above RM 10,000 must go through: (1) Direct Manager, (2) simultaneous approval from Department Head + Finance Controller, and (3) final sign-off by the COO.

SOLUTION: Use a combination of 'Start and wait' (Stage 1), two parallel 'Create approval' + 'Wait for approval' (Stage 2), and a final 'Start and wait' (Stage 3). Include a Condition at each stage — if any stage is rejected, immediately notify the requester and terminate the flow.

5.5 Approval Notifications via Outlook and Teams

Approvers receive notifications through two primary channels:

Channel	Approval Experience
Outlook Email	Approver receives email with Approve and Reject buttons embedded. No login required — decision made directly from inbox.
Microsoft Teams	Approval appears in the Teams Activity feed and Approvals app. Supports rich adaptive cards with comments and attachments.
Power Automate Approvals App	Dedicated approvals dashboard at flow.microsoft.com/approvals — shows all pending, completed, and sent approvals.

5.6 HR Leave Approval Workflow — Full Case Study

This end-to-end case study walks through a complete leave approval system:

Process Design

- Employee submits leave request via Microsoft Forms
- Request details stored in SharePoint 'Leave Requests' list
- Flow triggers on new SharePoint item
- Auto-checks leave balance from Excel tracker → if insufficient, rejects immediately
- Line Manager receives Teams approval card with employee details and leave dates
- If approved: HR receives notification | Leave balance deducted | Employee receives confirmation
- If rejected: Employee receives rejection with reason | Status updated in SharePoint



PARTICIPANT ACTIVITY

ACTIVITY 5A: Build a simplified Leave Approval Flow

Requirements: Form submission trigger → Manager approval → Employee notification

Time: 30 minutes | Use the SharePoint Leave Requests list prepared by the trainer

5.7 Finance Invoice Approval Workflow

Process Steps

82. Accounts Payable staff uploads invoice PDF to SharePoint document library
83. Flow triggers on file creation
84. AI Builder extracts invoice amount, vendor name, and invoice number from PDF
85. Condition: Amount > RM 5,000 → route to Finance Manager; Amount ≤ RM 5,000 → route to Team Lead
86. Approver receives Outlook email with invoice details and PDF link
87. If approved: Move file to 'Approved Invoices' folder | Update payment tracker in Excel
88. If rejected: Return to AP staff with rejection reason | Move to 'Pending Review' folder



MODULE 5 SUMMARY

- ✓ Power Automate Approvals connector manages full approval lifecycles
- ✓ Approval types: First to respond, Everyone must approve, Custom responses
- ✓ Sequential approvals process one approver at a time — mirrors corporate hierarchy
- ✓ Parallel approvals send to all approvers simultaneously — faster for peer-level decisions
- ✓ Approval notifications reach approvers via Outlook email and Microsoft Teams
- ✓ Real-world workflows combine sequential and parallel stages for complex governance

MODULE 6

Microsoft 365 Integration

Connecting Outlook, Teams, Excel, SharePoint, Forms & OneDrive

Module 6 — Microsoft 365 Integration

Power Automate's deepest integrations are with Microsoft 365 applications. This module provides comprehensive guidance on automating Outlook, Teams, Excel, SharePoint, Forms, and OneDrive — and combining them into cross-platform automation pipelines.

6.1 Outlook Automation

Outlook Trigger/Action	Use Case
When a new email arrives (V3)	Route emails, trigger notifications, auto-reply
When an email is flagged	Start a task creation workflow
Send an email (V2)	Automated notifications, reports, approval results
Reply to an email	Auto-acknowledge customer emails
Create event	Book calendar meetings from form submissions
Get calendar view of events	Check availability before scheduling

6.2 Teams Automation

Power Automate integrates with Microsoft Teams to enable automated messaging, channel management, and approval processes within the Teams interface.

- Post messages to channels — ideal for system alerts and project updates
- Post adaptive cards — interactive cards with buttons, dropdowns, and input fields
- Create Teams channels — automatically create project-specific channels on project initiation
- List team members — retrieve membership for notification routing
- @mention users — tag specific people in automated notifications

TRAINER DEMONSTRATION

Demonstrate creating an adaptive card in Teams that shows a KPI summary from SharePoint data

Show how to use 'Post adaptive card and wait for response' for in-Teams approvals

6.3 Microsoft Forms Integration

Microsoft Forms is commonly used for surveys, registrations, and data collection. Power Automate can trigger workflows from form submissions.

Tutorial: Forms to SharePoint Logger

89. Create a Microsoft Form with fields: Name, Department, Request Type, Description
90. In Power Automate: Create → Automated cloud flow
91. Trigger: 'When a new response is submitted' — Microsoft Forms
92. Form Id: Select your form from the dropdown
93. Add action: 'Get response details' — Forms | Form Id: [same form] | Response Id: [dynamic content]
94. Add action: 'Create item' — SharePoint | Map each form field to a SharePoint column
95. Add action: 'Send an email' — acknowledge submission to the respondent



INSERT SCREENSHOT HERE

Caption: Forms trigger with Get Response Details and SharePoint Create Item actions — dynamic content mapped from form fields

6.4 Cross-Platform Automation Example

End-to-End: Customer Feedback Pipeline

This example combines Forms, SharePoint, Teams, Outlook, and Excel:

- Customer submits feedback via Microsoft Forms
- Response stored in SharePoint Feedback list
- Sentiment analysed using AI Builder
- If negative: Alert posted in Teams #customer-success channel + manager email
- All responses logged to Excel dashboard for weekly review
- Automated weekly summary email sent every Friday at 5PM



MODULE 6 SUMMARY

- ✓ Outlook: Triggers on new/flagged emails; Actions include send, reply, create calendar events
- ✓ Teams: Post messages, adaptive cards, create channels, manage team membership
- ✓ Excel: Requires OneDrive/SharePoint storage + formatted Tables; supports all CRUD operations
- ✓ SharePoint: Full integration with Lists and Libraries — rich triggers and actions
- ✓ Forms: Trigger flows on submission; use 'Get response details' to extract field values
- ✓ Cross-platform flows combine multiple M365 connectors into unified business pipelines

MODULE 7

AI & Intelligent Automation

AI Builder, OCR, Document Processing & Sentiment Analysis

Module 7 — AI & Intelligent Automation

Power Automate integrates with AI Builder — Microsoft's AI platform — to bring intelligent capabilities such as document reading, image classification, language detection, and sentiment analysis into your automated workflows. No data science expertise is required.

7.1 AI Builder Introduction

AI Builder is a Power Platform capability that provides pre-built AI models and tools to add intelligence to your business processes.

AI Model Type	Description
Document Processing	Extracts structured data from forms, invoices, receipts, and contracts using machine learning
Object Detection	Identifies and counts objects in images — e.g., damaged goods, inventory counting
Text Classification	Categorises text into predefined labels — e.g., classify support tickets by category
Sentiment Analysis	Analyses text to determine positive, negative, or neutral sentiment
Language Detection	Identifies the language of input text
Business Card Reader	Extracts contact details from scanned business cards
Receipt Processing	Pre-built model for extracting common receipt fields (total, date, merchant)
Invoice Processing	Pre-built model for extracting invoice fields (vendor, amount, line items)

IMPORTANT NOTE

AI Builder requires a Power Automate Premium licence or AI Builder add-on credits. Pre-built models (Sentiment Analysis, Invoice Processing, Receipt Processing) work immediately without training.

Custom models (Document Processing, Object Detection) require training with your own sample documents.

7.2 OCR Automation — Reading Documents

Optical Character Recognition (OCR) converts printed or handwritten text in images and PDFs into machine-readable data that flows can process.

Tutorial: Process Uploaded Invoices Automatically

96. Create a flow triggered by: 'When a file is created' — SharePoint (Invoices folder)
97. Add action: 'Recognize text in an image or a PDF document' — AI Builder
98. Document: [File content from trigger]
99. Use the extracted text to check for key words — 'Total Amount', 'Invoice No.'
100. Route the invoice for approval based on the extracted amount

TRAINER NOTE

For Document Processing models, demonstrate in the AI Builder portal how to upload sample documents, label fields, and train the model before using it in a flow.

7.3 Sentiment Analysis Integration

Sentiment analysis automatically determines whether text (customer feedback, emails, survey responses) expresses a positive, negative, or neutral tone.

Tutorial: Auto-Escalate Negative Customer Feedback

101. Trigger: 'When a new response is submitted' — Microsoft Forms (Customer Feedback form)
102. Get response details including Comment field
103. Add AI Builder action: 'Detect the language being used in text'
104. Add AI Builder action: 'Analyze positive or negative sentiment in text'
105. Sentiment Text: [Comment from form response]
106. Add Condition: Sentiment is equal to 'negative'
107. YES: Post Teams alert to #customer-experience channel + flag SharePoint record
108. NO: Log response normally to SharePoint with 'Positive' or 'Neutral' tag

INSERT SCREENSHOT HERE

Caption: AI Builder Sentiment Analysis action in flow — showing Text input mapped from Forms response and Sentiment output

7.4 AI Use Cases in Workplace Automation

Business Function

AI Automation Use Case

Finance	AI invoice extraction → zero-touch invoice processing → auto-posting to accounting system
HR	Resume screening — classify uploaded CVs by suitability category before routing to recruiters
Customer Service	Analyse incoming emails → sentiment score → priority routing → auto-response drafting
Operations	Object detection on warehouse photos → automated inventory count updates
Legal & Compliance	Document classification — automatically tag contracts by type, department, expiry date



MODULE 7 SUMMARY

- ✓ AI Builder adds intelligent automation: document reading, sentiment analysis, text classification
- ✓ Pre-built AI models require no training — invoice, receipt, sentiment, language detection
- ✓ Custom models require sample documents for training before deployment
- ✓ OCR extracts text from images and PDFs for use in downstream flow actions
- ✓ Sentiment analysis enables automatic escalation of negative feedback
- ✓ AI Builder requires Power Automate Premium licence or AI Builder add-on credits

MODULE 8

Desktop Automation (RPA)

Power Automate Desktop & Robotic Process Automation

Module 8 — Desktop Automation (RPA)

Robotic Process Automation (RPA) enables automation of tasks that cannot be addressed through APIs alone — particularly legacy systems, desktop applications, and web-based portals that lack Power Automate connectors. Power Automate Desktop is Microsoft's built-in RPA tool.

8.1 Introduction to Power Automate Desktop

Power Automate Desktop is a free Windows application that enables you to create desktop flows — automated scripts that interact with your computer's interface just like a human user would.

Key capabilities:

- Record and replay user interface interactions (click, type, scroll, drag)
- Automate Windows desktop applications (Excel, Word, SAP, accounting software)
- Automate web browser interactions (Chrome, Edge, Firefox)
- Extract data from web pages (web scraping)
- Read and write files, folders, and registry entries
- Execute system commands and run PowerShell scripts

Power Automate Desktop is available free for Windows 10/11 users. Download from: [Microsoft Store](#) → 'Power Automate'



INSERT SCREENSHOT HERE

Caption: Power Automate Desktop application — showing the flow designer with Actions panel, Variables panel, and canvas

8.2 UI Automation

UI Automation interacts with Windows desktop applications by targeting specific UI elements — buttons, text fields, menus, and windows.

Common UI Automation Actions

- 'Click UI element' — click any button, link, or control in a Windows application
- 'Populate text field' — type text into any input field
- 'Select menu item' — navigate application menus

- 'Get details of a window' — retrieve window title, size, state
- 'Close window' — close any application window

UI Automation Best Practices

- Use the UI elements pane to capture and name UI elements — avoid relying on coordinates
- Add 'Wait for UI element' before interactions to handle variable loading times
- Use error handling for steps that may fail if UI elements are not found

8.3 Web Automation

Web automation enables Power Automate Desktop to control web browsers — navigating pages, filling forms, extracting data, and clicking buttons.

Web Action	Description
Launch new web browser	Open Chrome, Edge, or Firefox to a specified URL
Navigate to web page	Load a new URL in an already-open browser
Populate text field on web page	Enter text into a form field (login, search, data entry)
Click link on web page	Click any link, button, or element on a web page
Extract data from web page	Extract text, tables, or lists from a web page
Download file from web	Trigger file downloads from web applications

REAL-WORLD SCENARIO

SCENARIO: A finance team manually logs into a government portal every morning, downloads the daily exchange rate file, and updates their pricing spreadsheet. This takes 15 minutes daily.

DESKTOP FLOW SOLUTION: Power Automate Desktop launches Chrome, logs into the portal, navigates to the download page, downloads the Excel file, opens it, copies the exchange rate, opens the pricing spreadsheet, and updates the relevant cell — all in under 2 minutes, triggered by a scheduled cloud flow.

8.4 Desktop Recording

Power Automate Desktop includes a Recorder that captures your manual actions and converts them into flow steps automatically.

How to Record a Desktop Flow

109. Open Power Automate Desktop → Create a new flow
110. Click the 'Record' button in the toolbar
111. Perform the manual task as you normally would (click, type, navigate)

112. Click 'Done' when finished
113. Review and clean up the recorded actions — remove unnecessary steps
114. Parameterise hard-coded values with variables for reusability
115. Test the recording end-to-end

QUICK TIP

Recordings often capture extra steps (mouse movements, window resizing). Always review and remove unnecessary actions to make the flow more robust and maintainable.

8.5 Integrating Desktop Flows with Cloud Flows

Desktop flows are most powerful when called from cloud flows. This creates end-to-end automation that bridges the cloud and desktop:

116. Cloud flow receives a trigger (e.g., SharePoint item created, scheduled time reached)
117. Cloud flow gathers required data (e.g., form responses, Excel row values)
118. Cloud flow calls the desktop flow using 'Run a flow built with Power Automate Desktop'
119. Desktop flow receives input parameters and performs desktop automation
120. Desktop flow returns output values back to the cloud flow
121. Cloud flow continues with the returned data (notifications, updates, logging)

MODULE 8 SUMMARY

- ✓ Power Automate Desktop enables RPA — automating tasks that lack API connectors
- ✓ UI Automation interacts with Windows desktop application controls
- ✓ Web Automation controls browsers for form filling, navigation, and data extraction
- ✓ The Recorder captures manual actions and converts them into flow steps automatically
- ✓ Desktop flows integrate with cloud flows — cloud flow triggers → desktop flow executes → returns results
- ✓ Best for legacy systems (SAP, government portals, older ERP) without REST APIs

MODULE 9

Governance & Best Practices

Naming Conventions, Security, Monitoring & Enterprise Management

Module 9 — Governance & Best Practices

As organisations scale their automation programmes, governance becomes critical. This module covers the standards, security considerations, and management practices that ensure Power Automate deployments remain maintainable, secure, and compliant.

9.1 Naming Conventions

Consistent naming makes flows discoverable, understandable, and maintainable — especially as your organisation's automation portfolio grows.

Recommended flow naming format:

Format	Example
[DEPT] - [Trigger] - [Action] - [Version]	HR-LeaveRequest-ManagerApproval-v1
[DEPT] - [System] - [Purpose] - [Frequency]	FIN-SharePoint-InvoiceSync-Daily

9.2 Documentation Standards

Every production flow should include:

- Description: Explain what the flow does, why it exists, and what triggers it
- Owner: Name and email of the flow owner/maintainer
- Created date and last modified date
- Connection dependencies: List of connectors and accounts used
- Change log: Record of what changed and when

QUICK TIP

Use the 'Description' field in every flow (Edit → Flow details) and add a Comment action at the start of each major section within the flow canvas.

9.3 Security Considerations

- Use service accounts (dedicated Microsoft accounts) for production flows — never personal accounts, as the flow breaks if the individual leaves the company

- Follow the principle of least privilege — only grant connectors and permissions that are necessary
- Store credentials and API keys in Azure Key Vault — never hardcode them in flow actions
- Review and audit connections regularly — remove unused or orphaned connections
- Enable Multi-Factor Authentication (MFA) on all accounts used by production flows
- Restrict who can turn off or delete critical flows using SharePoint/Power Platform DLP policies

9.4 Environment Management

Environment	Purpose & Best Practice
Development	Build and test new flows; use dummy data; never use production credentials
Testing/UAT	User Acceptance Testing with near-real data; validate with actual business users
Production	Live flows serving real business processes; strict change control required

9.5 Monitoring and Analytics

Use Power Automate's built-in monitoring tools to track flow health:

- 'My Flows' → Click flow → 'Run history': Shows all runs with status, duration, and error details
- Power Automate Admin Centre (admin.powerplatform.microsoft.com): Organisation-wide flow analytics
- Power Platform Admin Centre → Analytics → Flow: Usage reports, connector activity, error rates
- Set up a monitoring flow: If any production flow fails → email admin team with flow name and error details

9.6 Performance Optimisation

- Use 'Run Only' permissions to share flows without exposing editing rights
- Limit Apply to Each concurrency to avoid throttling on connectors
- Use 'Filter Query' on SharePoint 'Get items' to retrieve only needed records — never retrieve all items and filter in the flow
- Use 'Top Count' to limit results when only a subset is needed
- Avoid chaining many sequential flows — consolidate into fewer, well-structured flows
- Schedule resource-intensive flows during off-peak hours (nights, weekends)



MODULE 9 SUMMARY

- ✓ Naming conventions: [DEPT]-[Trigger]-[Action]-[Version] ensures discoverability
- ✓ Every production flow needs a description, owner, connections list, and change log
- ✓ Security: Use service accounts, least privilege, Key Vault for credentials, MFA
- ✓ Maintain DEV → TEST/UAT → PRODUCTION environment progression

- ✓ Monitor flows via Run History, Admin Centre, and dedicated monitoring flows
- ✓ Optimise performance with filter queries, concurrency settings, and off-peak scheduling

MODULE 10**Real Business Case Studies***End-to-End Automation Architectures from HR, Finance & Operations*

Module 10 — Real Business Case Studies

This module examines complete, production-ready automation architectures based on common corporate scenarios. Each case study includes the business problem, solution design, flow architecture, connectors used, and expected outcomes.

10.1 HR Employee Onboarding Automation

Business Problem

Onboarding a new employee involves 8-12 manual steps across IT, HR, and Operations — taking up to 3 days to complete. Steps are often missed, causing delays in system access and a poor new joiner experience.

Automation Architecture

122. HR submits new employee form (Microsoft Forms: Name, Department, Start Date, Equipment needs)
123. Flow triggers on Forms submission
124. Create employee record in SharePoint 'Employee Directory' list
125. Send IT request via Teams to IT department channel (equipment, system access)
126. Create employee's OneDrive folder structure automatically
127. Add employee to relevant Teams channels based on department
128. Send welcome email to new employee with first-day information
129. Schedule reminder flow: 3 days before start → send checklist to line manager
130. Day 1: Auto-trigger orientation meeting invite in Outlook calendar

Connectors Used

- Microsoft Forms, SharePoint, Microsoft Teams, Outlook, OneDrive, Office 365 Groups

Expected Outcomes

- Onboarding time reduced from 3 days to 4 hours
- Zero missed steps — automated checklist ensures consistency
- Better new joiner experience — proactive, timely communications

10.2 Invoice Approval Automation — Finance

Architecture Summary

- 131. AP team uploads invoice to SharePoint Invoices library
- 132. AI Builder extracts: Vendor, Amount, Invoice Number, Date
- 133. Condition: Amount thresholds determine approval routing (Team Lead / Finance Manager / CFO)
- 134. Approver receives Outlook approval email with invoice link
- 135. On approval: Excel payment tracker updated | Invoice moved to Approved folder | AP team notified
- 136. On rejection: AP team notified with reason | Invoice moved to Rejected folder
- 137. Monthly: Scheduled flow generates payment summary report → sent to Finance Director

10.3 Customer Support Ticket Automation

Architecture Summary

- 138. Customer submits support request via Microsoft Forms or email
- 139. AI Builder classifies request category (Billing, Technical, General)
- 140. Sentiment analysis determines priority (Negative = High, Neutral = Medium, Positive = Low)
- 141. SharePoint ticket list item created with all extracted details
- 142. Relevant support team notified in Teams with ticket summary
- 143. Auto-acknowledgement sent to customer within 2 minutes of submission
- 144. SLA timer triggered: if no update after 4 hours → escalate to supervisor
- 145. Weekly SLA performance report generated and sent to Customer Success Manager

10.4 Daily Reporting Automation

Architecture Summary

- 146. Scheduled trigger: Every weekday at 7:30 AM
- 147. Retrieve previous day's data from SharePoint lists (Sales, Tickets, Approvals)
- 148. Calculate KPIs using expressions: total, count, average, % change
- 149. Build HTML table summary
- 150. Send formatted report email to leadership team
- 151. Post condensed KPI summary card to Teams #daily-briefing channel



MODULE 10 SUMMARY

- ✓ Real automation architectures combine multiple connectors and flow patterns
- ✓ HR onboarding: Forms → SharePoint → Teams → Outlook → OneDrive in one coordinated flow
- ✓ Finance: AI document extraction + conditional routing + Excel updating + reporting
- ✓ Customer support: Forms/email → AI classification → SharePoint → Teams → SLA management

✓ Daily reporting: Scheduled trigger → data retrieval → KPI calculation → email/Teams delivery

MODULE 11

Hands-On Labs

Guided Step-by-Step Practical Exercises

Module 11 — Hands-On Labs

This module contains six fully guided laboratory exercises. Each lab builds on concepts from previous modules and results in a working automated flow that participants can adapt for their own workplace needs.

LAB 1: Build an Approval Flow

LAB OVERVIEW

Objective: Create a document approval workflow

Duration: 30 minutes

Level: Intermediate

Connectors: SharePoint, Approvals, Outlook

Instructions

152. Create a SharePoint list named 'Document Requests' with columns: Title (text), Requester (text), Document Type (choice), Status (choice: Pending/Approved/Rejected)
153. Create → Automated cloud flow | Trigger: 'When an item is created' — SharePoint | Site/List: Document Requests
154. Add: 'Start and wait for an approval' | Type: Approve/Reject — First to respond
155. Assigned To: [enter your manager's email or a colleague for testing]
156. Title: 'Document Approval: [Title from SharePoint]' | Details: 'Requester: [Requester] | Document Type: [Document Type]'
157. Add Condition: Outcome is equal to 'Approve'
158. YES: Update item → Status: Approved | Send email to Requester: 'Your request has been approved.'
159. NO: Update item → Status: Rejected | Send email to Requester: 'Your request has been rejected.'
160. Save → Test → Create a test item in the SharePoint list → Confirm email received



PARTICIPANT ACTIVITY

Extension challenge: Add a comment field to the approval that gets included in the notification email.

Advanced: Add a second approver (HR Manager) in sequence after the first approval.

LAB 2: Save Email Attachments Automatically

LAB OVERVIEW

Objective: Auto-save email attachments to SharePoint

Duration: 20 minutes

Level: Beginner-Intermediate

Connectors: Outlook, SharePoint

161. Create → Automated cloud flow | Trigger: 'When a new email arrives (V3)'
162. Configure: Has Attachments = Yes | Subject Filter: 'Report' (optional)
163. Add: 'Apply to each' | Select output: Attachments
164. Inside loop: Add 'Create file' — SharePoint
165. Site: [Your SharePoint site] | Folder: /Shared Documents/Email Attachments
166. File Name: [Attachment Name] | File Content: [Attachment Content]
167. Save → Test by sending yourself an email with 'Report' in subject and a PDF attached
168. Verify file appears in SharePoint within 2 minutes

LAB 3: Send Teams Notifications from SharePoint

LAB OVERVIEW

Objective: Post Teams messages when SharePoint list items are created

Duration: 20 minutes

Level: Beginner

Connectors: SharePoint, Microsoft Teams

169. Create → Automated cloud flow | Trigger: 'When an item is created' — SharePoint
170. Select a SharePoint list (e.g., Tasks, Requests, or Issues)
171. Add: 'Post a message in a chat or channel' — Microsoft Teams
172. Post as: Flow bot | Post in: Channel | Team: [Select your team] | Channel: General
173. Message: 'New item added: [Title] | Submitted by: [Created By DisplayName] | Date: [Created]'
174. Save → Test → Create a new item in the SharePoint list → Verify Teams message appears

LAB 4: Update Excel Automatically from Forms

LAB OVERVIEW

Objective: Log Microsoft Forms responses into an Excel table

Duration: 25 minutes

Level: Beginner-Intermediate

Connectors: Forms, Excel Online, Outlook

175. Create a Microsoft Form: 'Staff IT Request' — fields: Name, Department, Request Type, Description, Urgency
176. Create an Excel file in OneDrive named 'IT_Requests_Log.xlsx' — create a Table with matching columns
177. Create → Automated cloud flow | Trigger: 'When a new response is submitted' — Forms
178. Add: 'Get response details' — Forms | Response Id: [dynamic content]
179. Add: 'Add a row into a table' — Excel Online (Business)
180. Location: OneDrive for Business | File: IT_Requests_Log.xlsx | Table: Table1
181. Map each Excel column to the corresponding form field from dynamic content
182. Add: 'Send an email (V2)' — Outlook | To: form respondent's email | Subject: 'IT Request Received' | Body: Confirmation message
183. Save → Test → Submit the form → Check Excel and email

LAB 5: Create a Scheduled Reminder Flow

LAB OVERVIEW

Objective: Send automated weekly reminders via email and Teams

Duration: 20 minutes

Level: Beginner

Connectors: Outlook, Microsoft Teams

184. Create → Scheduled cloud flow | Name: 'Weekly Team Reminder' | Start: [today] | Repeat every: 1 Week on Monday
185. Add: 'Initialize variable' | Name: weekNumber | Type: String | Value: formatDateTime(utcNow(), 'WW')
186. Add: 'Send an email (V2)' — Outlook | To: [team distribution email]
187. Subject: 'Weekly Reminder — Week [weekNumber variable]'
188. Body: 'Reminder: Please submit your weekly report by Wednesday 5PM. Questions? Contact your line manager.'
189. Add: 'Post a message in a chat or channel' — Teams | Post the same reminder to the team channel
190. Save → Test manually to verify email and Teams message are sent correctly

LAB 6: Build a Leave Approval System

LAB OVERVIEW

Objective: Full end-to-end leave request and approval workflow

Duration: 40 minutes

Level: Advanced

Connectors: Forms, SharePoint, Approvals, Outlook, Teams

191. Create a Microsoft Form: 'Leave Application' — fields: Employee Name, Email, Leave Type, Start Date, End Date, Reason
192. Create a SharePoint list: 'Leave Applications' — columns matching form fields + Status column
193. Create → Automated cloud flow | Trigger: 'When a new response is submitted' — Forms
194. Get response details → Create SharePoint item with all form fields + Status: Pending
195. Add: 'Start and wait for an approval' | Assigned to: [manager email] | Title: 'Leave Application: [Employee Name]'
196. Details: 'Leave Type: [Type] | Dates: [Start] to [End] | Reason: [Reason]'
197. Add Condition: Outcome equals 'Approve'
198. YES: Update SharePoint status → Approved | Send confirmation email to employee | Post Teams notification
199. NO: Update SharePoint status → Rejected | Send rejection email with manager's comments
200. Save → Test end-to-end with a test form submission

TRAINER NOTE

Allow participants to help each other during Lab 6 — collaborative learning is encouraged
Trainer circulates the room and provides 1-on-1 assistance for participants who are stuck
Extension: Add an automatic rejection if leave balance is insufficient (check Excel leave balance table)

MODULE 12

Troubleshooting & Common Issues

Diagnosing, Debugging & Resolving Flow Problems

Module 12 — Troubleshooting & Common Issues

Even well-designed flows encounter issues. This module equips participants with systematic troubleshooting skills to diagnose and resolve the most common Power Automate problems encountered in corporate environments.

12.1 Failed Flows — Diagnosis Approach

Follow this systematic approach when a flow fails:

201.

Navigate to My Flows → Click the flow name → Click 'Run history' tab
202.

Identify the failed run → Click on it to open the run details
203.

Look for the red X icon — this indicates the first failed step
204.

Click the failed step to expand it → Review the Inputs and Outputs
205.

Read the error message carefully — note the error code and description
206.

Check if the issue is with inputs (wrong dynamic content), connection (authentication), or logic (wrong condition)

 INSERT SCREENSHOT HERE

Caption: Run history panel showing a failed flow run — red X on the failing step with error message expanded

12.2 Authentication & Connection Problems

Error	Cause & Solution
'Connection not authenticated'	The connection account's password changed or MFA was reset. Go to Data → Connections → fix the flagged connection by re-authenticating.
'401 Unauthorized'	Insufficient permissions on the target service (SharePoint site, mailbox). Ensure the connection account has appropriate permissions.
'403 Forbidden'	The account does not have access to the resource. Check SharePoint permissions,

	Exchange mailbox permissions, or Teams membership.
'Invalid client secret'	An API key or client secret has expired. Update credentials in the custom connector or Azure Key Vault.

12.3 Connector Limitations & Throttling

Each connector has API limits that govern how many requests can be made per minute, hour, or day. Exceeding these limits causes 429 'Too Many Requests' errors.

- SharePoint: 600 requests per minute per connection
- Outlook: 300 requests per minute per connection
- Teams: 30 requests per second per connection

Solutions for Throttling

- Reduce concurrency in Apply to Each (Settings → Concurrency Control)
- Add a 'Delay' action between iterations to slow down request rates
- Use 'Filter Query' on SharePoint Get Items to reduce the number of items retrieved
- Batch operations where possible instead of individual item-by-item processing

12.4 Timeout Errors

Flows have a maximum duration of 30 days. Individual actions have timeouts of 2 hours for synchronous calls. Solutions:

- Break large processes into smaller flows called sequentially
- Use asynchronous patterns — 'Create approval' + 'Wait for approval' instead of 'Start and wait'
- Move data-intensive processing to Power BI dataflows or Azure Data Factory

12.5 Dynamic Content Issues

Issue	Solution
Dynamic content not available	The step producing the content may be inside a different branch. Verify the content source is in the main flow path before your current step.
'Expression is invalid'	Syntax error in expression. Check for missing quotes, wrong function name, or incorrect parameter count using the Expression editor.
Null value error	A dynamic content field is empty. Add a Condition or use the coalesce() expression to handle nulls: coalesce(field, 'default value')
Wrong data type	A text field is expected but a date is provided. Use formatDate() or string() to convert types.

12.6 Monitoring Flow Runs

- Set up a dedicated 'Flow Monitor' flow: Query Power Automate's management API for failed flows → send daily email summary to the automation owner
- Use Power Platform Admin Centre → Analytics → Power Automate for organisation-wide monitoring
- Configure email alerts in 'My Flows' → Flow detail → notification settings
- For critical flows: Create a SharePoint log list where the flow writes its own status after each run

12.7 Debugging Techniques

- Use 'Compose' action as a debug probe — output any value or expression to see what it actually contains
- Rename actions clearly so run history is easy to read
- Test with static values first, then switch to dynamic content once the flow logic is confirmed
- Use the 'Flow checker' (before saving) to catch configuration errors early
- Break complex flows into smaller sub-flows for easier isolated testing



MODULE 12 SUMMARY

- ✓ Always start troubleshooting with Run History — identify the exact failed step
- ✓ Connection errors: re-authenticate via Data → Connections in Power Automate
- ✓ Throttling: reduce concurrency, add delays, use filter queries
- ✓ Dynamic content issues: check null values, data types, and expression syntax
- ✓ Use Compose actions as debug probes to inspect values in the flow
- ✓ Monitor production flows proactively — don't wait for users to report failures

MODULE 13**Final Project***Build a Complete Business Automation Solution*

Module 13 — Final Project

The Final Project is the capstone of this training programme. Participants will design, build, test, and present a complete business automation solution using Microsoft Power Automate — demonstrating mastery of the concepts covered across all modules.

13.1 Project Brief

You have been engaged as an automation consultant for an organisation of your choice (real or fictional). Your task is to identify a real business problem in that organisation and design a complete end-to-end automated solution using Power Automate.

13.2 Suggested Scenarios

- Staff Performance Review Automation: Form submission → approval chain → HR notification → SharePoint logging → Outlook confirmation
- Supplier Invoice Processing: File upload → AI extraction → approval routing → Excel update → payment notification
- IT Helpdesk Ticket System: Forms submission → AI classification → Teams alert → SharePoint ticket → SLA reminder → resolution confirmation
- Event Registration & Confirmation: Forms → approval (for paid events) → calendar invite → Teams channel → reminder 24 hours before event
- Monthly KPI Report Distribution: Scheduled trigger → SharePoint data retrieval → KPI calculation → HTML report → email to stakeholders
- Custom scenario of your choice: Approved by trainer before development begins

13.3 Requirements Checklist

Requirement	Description
Flow Trigger	At least one trigger — Forms, SharePoint, Schedule, or Instant
Approval Component	At least one approval step with conditional handling of outcomes
Condition/Logic	At least two conditions or logical branches
Variable Usage	At least one variable used within the flow

Microsoft 365 Integration	Integration with at least three M365 services
Error Handling	At least one 'Configure run after' or scope-based error handler
Notification	Send at least one email AND one Teams message as outcome notifications
Documentation	Flow description, naming convention, and connection list documented

13.4 Evaluation Criteria

Criteria	Weightage	Excellent (4)	Needs Work (1-2)
Technical Accuracy	30%	Flow runs end-to-end without errors	Multiple errors; flow does not complete
Business Relevance	20%	Solves a real, meaningful business problem	Solution is trivial or hypothetical
Complexity & Coverage	20%	Uses 5+ modules' concepts effectively	Basic flow with minimal integration
Presentation	15%	Clear, confident, well-structured walkthrough	Unclear explanation; skips key steps
Documentation	15%	Complete naming, description, and change log	No documentation; non-standard naming

13.5 Presentation Guidelines

207. Duration: 10-12 minutes per participant/group
208. Start with: Business problem statement — what are you solving and why?
209. Show: Flow diagram or architecture overview (hand-drawn or digital)
210. Demonstrate: Live demo of the complete flow running end-to-end
211. Explain: Key design decisions — why did you choose this approach?
212. Conclude: What is the estimated time/cost saving? What would you add next?

TRAINER NOTE

Final project presentations can be conducted on Day 2 afternoon (1.5 hours for a group of 10-15)

Encourage participants to select scenarios from their actual workplace — this maximises immediate applicability

Provide 15-20 minute structured build time during training for participants to progress their project

Conclusion & Course Recap

Congratulations on completing the Microsoft Power Automate: Workflow Automation for Workplace Productivity training programme. You have covered a comprehensive journey from foundational automation concepts to advanced enterprise-grade workflow design.

Key Takeaways

☑ WHAT YOU CAN NOW DO

Build automated cloud flows for email, SharePoint, Teams, Excel, and Forms
Design sequential, parallel, and multi-stage approval workflows
Use conditions, variables, loops, and expressions to build intelligent automation logic
Integrate AI Builder for document processing, sentiment analysis, and OCR
Automate desktop and legacy applications using Power Automate Desktop (RPA)
Apply governance best practices: naming conventions, documentation, security, monitoring
Analyse business processes and design end-to-end automation architectures

Microsoft Certification Roadmap

Power Automate skills align with several Microsoft Power Platform certifications:

Certification	Description & Target Audience
PL-900: Microsoft Power Platform Fundamentals	Beginner-level certification covering all Power Platform components. Ideal starting point for participants new to Power Platform. Recommended: take within 2 months of this training.
PL-200: Microsoft Power Platform Functional Consultant	Intermediate certification focusing on Power Apps and Power Automate implementation for business consultants. Requires 1-2 years of hands-on experience.
PL-500: Microsoft Power Automate RPA Developer	Advanced certification specifically for Power Automate Desktop and RPA development. Covers attended/unattended automation, deployment, and governance.

Future Learning Roadmap

- Week 1-2: Revise all lab exercises; adapt at least one flow for your actual workplace
- Week 3-4: Explore Microsoft Learn Power Automate learning paths (learn.microsoft.com)
- Month 2: Begin PL-900 exam preparation using official Microsoft study materials
- Month 3-4: Identify 3-5 automation opportunities in your organisation and implement them

- Month 5-6: Explore AI Builder custom model training; expand into Power Apps integration
- Month 6+: Consider PL-200 or PL-500 certification based on your specialisation focus

Additional Learning Resources

Resource	Type	URL/Location	Best For
Microsoft Learn	Free online platform	learn.microsoft.com/power-automate	Self-paced learning paths and certifications
Power Automate Community	Forum & Blog	powerusers.microsoft.com	Community Q&A, shared templates, tips
Power Automate YouTube	Video tutorials	youtube.com/PowerAutomate	Visual walkthroughs of features
Power Automate Blog	Official updates	powerautomate.microsoft.com/blog	New features and announcements
Microsoft Docs	Technical reference	docs.microsoft.com/power-automate	Detailed connector documentation

Q&A Session

DISCUSSION QUESTIONS

Q: Which automation from today's training could you implement at your workplace this week?

Q: What business process in your organisation wastes the most time that could be automated?

Q: What support do you need from management or IT to begin implementing Power Automate?

Q: Which certification from the roadmap aligns best with your career goals?

STAY CONNECTED

Dr. Fakhrul Syahmi | Data Consultant & AI Trainer

Email: mufasyahcons@gmail.com

For post-training support, custom automation consulting, or follow-up training enquiries — please reach out via email.

Thank you for your participation and enthusiasm throughout this training programme. You now have the skills to transform your organisation through intelligent automation.

Glossary of Key Terms

Term	Definition
Action	A step in a Power Automate flow that performs an operation, such as sending an email or creating a SharePoint item.
Adaptive Card	An interactive card format used in Microsoft Teams and Outlook that can contain buttons, input fields, and formatted content.
AI Builder	A Microsoft Power Platform capability providing pre-built and custom AI models for use in Power Automate flows.
Attended Automation	RPA flows that run with a human user present and logged into the machine.
Cloud Flow	A Power Automate flow that runs in the cloud and connects to online services via APIs.
Concurrency	The ability to process multiple iterations of a loop simultaneously (in parallel) rather than one at a time.
Connector	An interface that allows Power Automate to communicate with external services and applications.
Dynamic Content	Variable data from previous flow steps that can be inserted into subsequent step configurations.
Environment	A container in Power Platform that groups flows, data, and resources, separated by purpose (DEV, TEST, PROD).
Expression	A formula using built-in functions to manipulate data within Power Automate (e.g., formatDateTime, concat).
Flow	An automated workflow created in Power Automate consisting of a trigger and one or more actions.
Low-Code	A development approach requiring minimal hand-coding, enabling non-developers to build solutions through visual interfaces.
OCR	Optical Character Recognition — technology that converts text in images or PDFs into machine-readable data.
Power Automate Desktop	Microsoft's desktop RPA tool that enables automation of local applications and web browsers.

Power Platform	Microsoft's suite of low-code tools including Power BI, Power Apps, Power Automate, Power Pages, and Copilot Studio.
Premium Connector	A connector requiring a paid Power Automate Premium licence, providing access to non-Microsoft services.
RPA	Robotic Process Automation — technology that uses software robots to mimic human interactions with computer systems.
Run History	A log of all past and current executions of a Power Automate flow, showing status, duration, and error details.
Scope	A container action in Power Automate that groups steps together, enabling error handling for the entire group.
Sequential Approval	An approval pattern where each approver is notified in order — the next only receives the request after the previous has responded.
Standard Connector	A connector included with Microsoft 365 subscriptions, accessing Microsoft services without additional licencing.
Trigger	The event that starts a Power Automate flow — occurs once at the beginning of every flow.
Unattended Automation	RPA flows that run without a human user present — typically on a virtual machine or server.
Variable	A named storage location in a flow that holds a value (text, number, boolean, or array) that can change during execution.
Webhook	An HTTP callback mechanism that sends real-time data from one system to another when an event occurs.